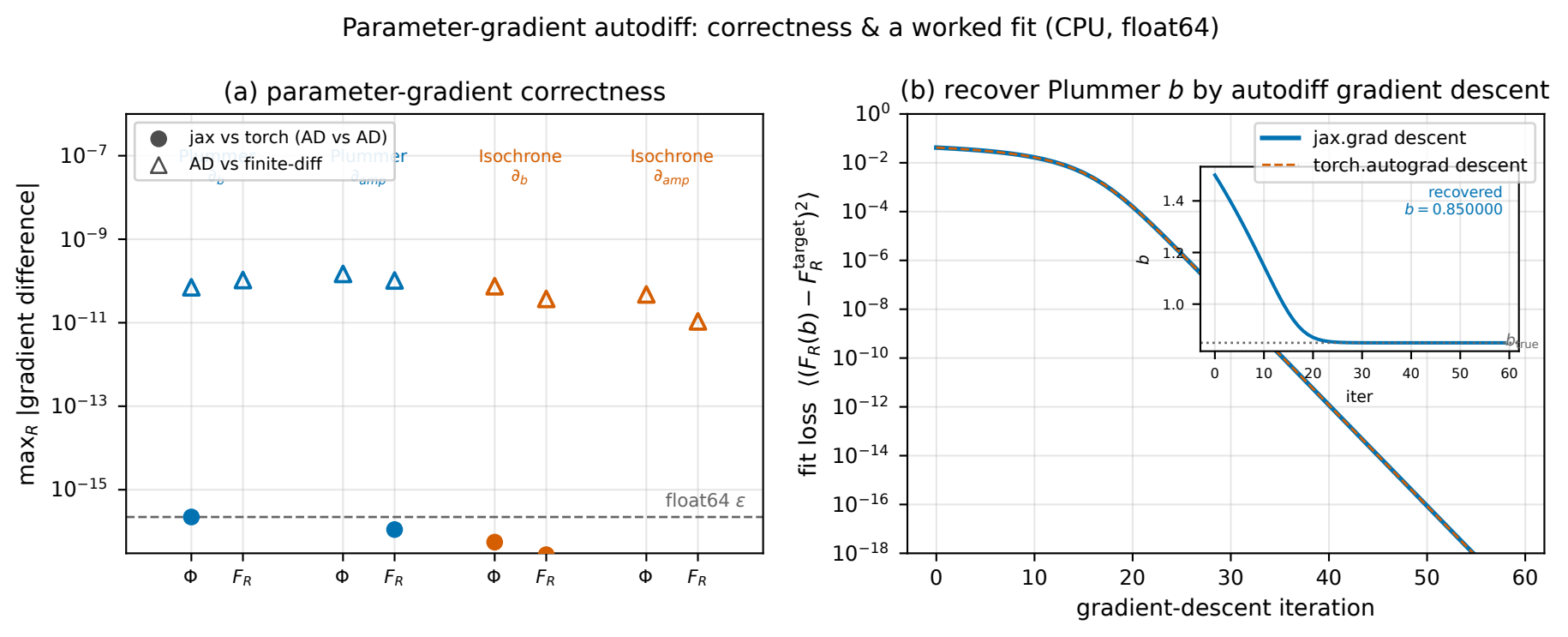
Parameter-gradient autodiff: correctness & a worked fit (CPU, float64)
(a) parameter-gradient correctness
jax vs torch (AD vs AD)
AD vs finite-diff
Plummer ∂b
Plummer ∂amp
Isochrone ∂b
Isochrone ∂amp
float64 ε
max_R |gradient difference|
Φ
F_R
(b) recover Plummer b by autodiff gradient descent
jax.grad descent
torch.autograd descent
recovered
b = 0.850000
b_true
iter
fit loss ⟨(F_R(b) − F_R^target)²⟩
gradient-descent iteration